

Lecture 2 — From Values to Demand

Lecturer: Sid Banerjee

Date: August 27, 2026

2.1 Overview

In Lecture 1 we assumed that every buyer's value was known. This let us ask what prices can accomplish when information is not the issue: divide gains from trade, allocate scarce supply, coordinate decentralized choices, and manage dynamic opportunity costs or congestion externalities.

We now move to a setting with *partial information*. The seller no longer knows an individual buyer's realized value, but knows the distribution from which values are drawn. Our central question is:

How does a distribution of buyer values generate demand and revenue?

We will first separate random sales and revenue from their expectations. We will then discover that a change from price to market-share coordinates exposes useful economic structure. Along the way we will connect four objects that recur throughout the course:

elasticity, hazard rate, virtual value, choice probability.

Sections marked * are optional reading. They deepen this development but are not needed for the core chain of ideas.

2.2 From reservation values to a demand curve

Begin with a single potential buyer whose value is a random variable

$$V \sim F.$$

The seller knows the cumulative distribution function (CDF)

$$F(x) = \mathbb{P}(V \leq x),$$

but not the realization of V when choosing a price. Thus $F(x)$ records the probability mass at or below each possible threshold x . We assume F is continuous throughout the core development.

There are two useful interpretations of this model.

- (1) **Population interpretation.** A buyer is sampled from a large population whose values have distribution F .
- (2) **Behavioral interpretation.** The same person's value varies with unobserved tastes or circumstances; V is the random component of a utility model.

The mathematics is the same in either case. The second interpretation will reappear in the optional random-utility section.

2.2.1 One buyer: random demand and random revenue

At a posted price p , the buyer purchases if and only if

$$V \geq p.$$

The realized demand is therefore the Bernoulli random variable

$$Q(p) = \mathbf{1}\{V \geq p\} \sim \text{Bernoulli}(1 - F(p)).$$

Writing

$$q(p) := 1 - F(p),$$

we have

$$\mathbb{E}[Q(p)] = q(p).$$

With zero marginal cost, realized revenue and expected revenue per potential customer are

$$R(p) = pQ(p), \tag{2.1}$$

$$r(p) := \mathbb{E}[R(p)] = p(1 - F(p)) = pq(p). \tag{2.2}$$

The distinction matters: $R(p)$ is random, whereas $r(p)$ is a deterministic function that the seller can optimize.

For reference, the corresponding variances are

$$\text{Var}(Q(p)) = F(p)(1 - F(p)), \quad \text{Var}(R(p)) = p^2 F(p)(1 - F(p)).$$

2.2.2 Many independent buyers: aggregate demand

Now suppose there are n potential buyers with independent values

$$V_1, \dots, V_n \stackrel{\text{iid}}{\sim} F.$$

At price p , realized aggregate demand is

$$Q_n(p) = \sum_{i=1}^n \mathbf{1}\{V_i \geq p\} \sim \text{Binomial}(n, 1 - F(p)). \tag{2.3}$$

Its expectation is the deterministic demand curve

$$d(p) := \mathbb{E}[Q_n(p)] = n\mathbb{P}(V_i \geq p) = n(1 - F(p)). \quad (2.4)$$

This is the simplest bridge from micro-level heterogeneity to an aggregate demand curve.

Realized total revenue and its expectation are

$$R_n(p) = pQ_n(p), \quad (2.5)$$

$$r_n(p) := \mathbb{E}[R_n(p)] = np(1 - F(p)) = nr(p). \quad (2.6)$$

For most of this lecture we work with the per-customer expected revenue

$$r(p) = p(1 - F(p)).$$

The factor n can always be restored at the end.

	Random quantity	Deterministic expectation
One buyer	$Q(p), R(p)$	$q(p) = \mathbb{E}[Q(p)], r(p) = \mathbb{E}[R(p)]$
n buyers	$Q_n(p), R_n(p)$	$d(p) = \mathbb{E}[Q_n(p)], r_n(p) = \mathbb{E}[R_n(p)]$

Remark 2.1 (What changes in a large market?). *Independence gives*

$$\begin{aligned} \text{Var}(Q_n(p)) &= nF(p)(1 - F(p)), \\ \text{Var}(R_n(p)) &= np^2F(p)(1 - F(p)). \end{aligned}$$

For a fixed purchase probability in $(0, 1)$, expected demand and revenue are of order n , while their standard deviations are of order $\sqrt{n}$. Their relative fluctuations are therefore of order $n^{-1/2}$.

First-order quantities scale like n , ordinary statistical fluctuations scale like $\sqrt{n}$.

Informally, the central limit theorem says that a sum of many independent, similarly distributed terms looks approximately Gaussian after centering and scaling:

$$\sum_{i=1}^n X_i \approx n\mu + \sqrt{n}\sigma Z, \quad Z \sim \mathcal{N}(0, 1),$$

when the X_i are i.i.d. with mean μ and variance σ^2 . The same n versus $\sqrt{n}$ comparison will recur later when a fluid benchmark has scale T but stochastic deviations have scale $\sqrt{T}$.

2.3 Price space is not always the nicest coordinate system

We now ask whether price p is the most revealing coordinate in which to optimize

$$r(p) = p(1 - F(p)).$$

A natural hope is that $r(p)$ is concave in p , so that the pricing problem is a standard concave maximization problem. This fails even for extremely familiar distributions.

Example 2.2 (Exponential values). *Let*

$$V \sim \text{Exp}(1), \quad F(p) = 1 - e^{-p}, \quad p \geq 0.$$

Then

$$d(p) = ne^{-p}$$

and

$$r(p) = pe^{-p}.$$

Differentiating,

$$r'(p) = e^{-p}(1 - p),$$

so the optimal price is $p^ = 1$. But*

$$r''(p) = e^{-p}(p - 2).$$

Hence r is concave for $p < 2$ and convex for $p > 2$. Even this very smooth, unimodal pricing problem is not globally concave in price.

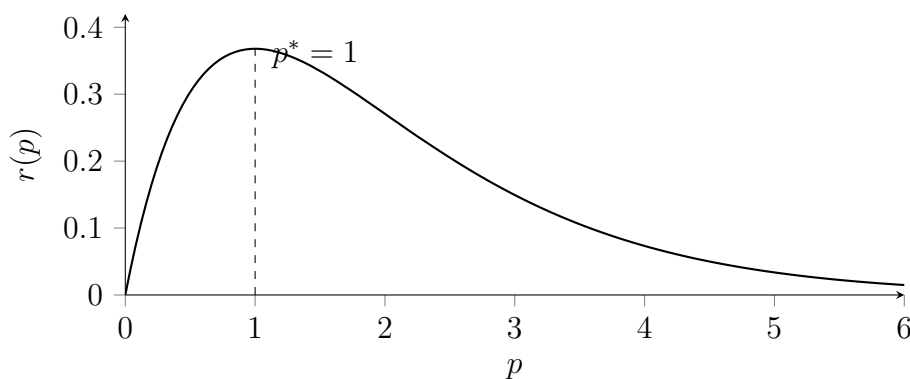


Figure 2.1. For exponential values, revenue pe^{-p} is easy to optimize but is not globally concave in the price p .

This suggests a change of variables.

2.4 Quantile space: choose market share instead of price

Define the *sale probability* or *demand quantile*

$$q = q(p) = 1 - F(p). \quad (2.7)$$

If F is continuous and strictly increasing, we can invert this relation:

$$p = v(q) := F^{-1}(1 - q), \quad q \in [0, 1]. \quad (2.8)$$

Instead of thinking of the seller as choosing a price, we can think of the seller as choosing the fraction q of the market she wants to serve. As q rises, the seller serves buyers with progressively lower values. The corresponding per-customer revenue is the *revenue curve*

$$r(q) = qv(q) = qF^{-1}(1 - q). \quad (2.9)$$

This is the same expected-revenue function as $r(p) = p(1 - F(p))$, written in a different coordinate. We use $r(p)$ when the argument is price and $r(q)$ when the argument is market share. Total expected revenue from n potential buyers is $nr(q)$.

Example 2.3 (Exponential values, revisited). For $V \sim \text{Exp}(1)$,

$$q = e^{-p}, \quad p = -\log q.$$

Therefore

$$r(q) = -q \log q.$$

Now

$$r''(q) = -\frac{1}{q} < 0,$$

so revenue is concave in quantile space.

The transformation does not magically make every distribution well behaved. Rather, it identifies the right structural condition under which revenue is concave.

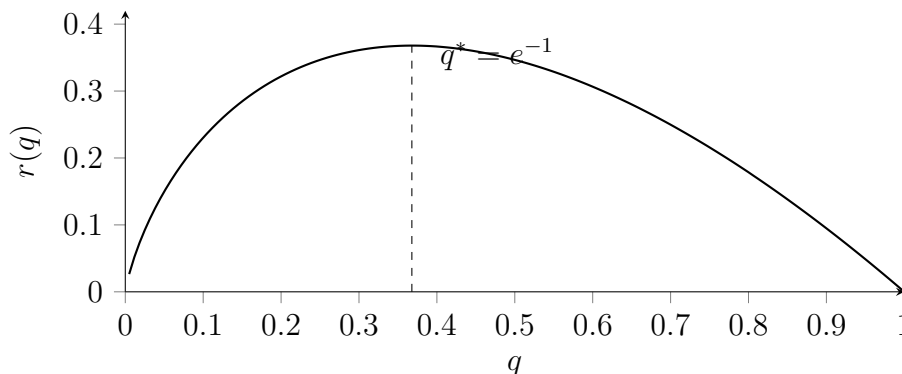


Figure 2.2. The same exponential pricing problem becomes a concave maximization problem in the demand quantile q .

2.5 Marginal revenue and virtual value

Assume F has a positive density f on the region of interest. Since

$$q = 1 - F(v),$$

we have

$$\frac{dq}{dv} = -f(v) \quad \implies \quad \frac{dv}{dq} = -\frac{1}{f(v)}.$$

Differentiating the revenue curve gives

$$r'(q) = v(q) + qv'(q) \tag{2.10}$$

$$= v - \frac{1 - F(v)}{f(v)}. \tag{2.11}$$

This motivates an important definition.

Definition 1 (Virtual value / marginal revenue). For a value distribution F with density f , define

$$\phi(v) := v - \frac{1 - F(v)}{f(v)}. \tag{2.12}$$

Then

$$\boxed{r'(q) = \phi(v(q))}. \tag{2.13}$$

For the moment, there is no auction theory hiding here: $\phi(v)$ is simply the *marginal revenue from increasing the served market quantile*. We will encounter exactly the same function much later when we study optimal auctions.

2.5.1 Revenue regularity

Because $v(q)$ is decreasing in q , the derivative $r'(q) = \phi(v(q))$ is decreasing in q exactly when $\phi(v)$ is increasing in v .

Definition 2 (Regular value distribution). A distribution F is regular if its virtual value

$$\phi(v) = v - \frac{1 - F(v)}{f(v)}$$

is nondecreasing in v .

Equivalently,

$$F \text{ regular} \iff r(q) = qF^{-1}(1 - q) \text{ is concave in } q. \quad (2.14)$$

Remark 2.4 (Terminology). Talluri–van Ryzin also use the word “regularity” for a collection of smoothness and boundedness assumptions on an aggregate demand function. Here we use regular distribution in the mechanism-design sense: monotone virtual value, equivalently concavity of the quantile revenue curve.

Example 2.5 (Uniform values). If $V \sim \text{Unif}[0, 1]$, then

$$1 - F(v) = 1 - v, \quad f(v) = 1,$$

so

$$\phi(v) = 2v - 1.$$

Hence F is regular. In quantile space, $v(q) = 1 - q$ and

$$r(q) = q(1 - q),$$

a concave quadratic.

2.6 Hazard rates, virtual values, and elasticity

The hazard rate of F is

$$h(v) := \frac{f(v)}{1 - F(v)}. \quad (2.15)$$

Hence virtual value can be written as

$$\phi(v) = v - \frac{1}{h(v)}. \quad (2.16)$$

If the hazard rate is nondecreasing, then $1/h(v)$ is nonincreasing, so $\phi(v)$ is strictly increasing. Thus

$$\boxed{\text{monotone hazard rate (MHR)} \implies \text{regular.}} \quad (2.17)$$

The converse need not hold.

Now return to deterministic aggregate demand

$$d(p) = n(1 - F(p)).$$

Its (absolute) price elasticity is

$$\epsilon(p) := -\frac{pd'(p)}{d(p)}. \quad (2.18)$$

Since $d'(p) = -nf(p)$,

$$\boxed{\epsilon(p) = \frac{pf(p)}{1 - F(p)} = ph(p).} \quad (2.19)$$

These objects all encode the same local tradeoff. Differentiating per-customer expected revenue,

$$r'(p) = (1 - F(p)) - pf(p) \quad (2.20)$$

$$= (1 - F(p)) [1 - \epsilon(p)]. \quad (2.21)$$

Thus an interior revenue-maximizing price p^* satisfies

$$\boxed{\epsilon(p^*) = 1.} \quad (2.22)$$

Using $\epsilon(p) = ph(p)$ and (2.16), this is equivalent to

$$\boxed{\phi(p^*) = 0.} \quad (2.23)$$

So, for an interior optimum, the same necessary first-order condition can be read three ways:

$$\boxed{\text{interior pricing FOC} \iff \text{unit elasticity} \iff p^*h(p^*) = 1 \iff \phi(p^*) = 0.}$$

Example 2.6 (Exponential values, one last time). For $V \sim \text{Exp}(1)$,

$$h(v) = 1, \quad \phi(v) = v - 1, \quad \epsilon(p) = p.$$

All three views give $p^* = 1$ immediately.

Next lecture we will add marginal cost c . The zero-virtual-value condition then becomes $\phi(p) = c$, which is the distributional version of the familiar monopoly markup rule.

2.7 Optional: mixtures and nonregularity^{*}

Regularity is useful, but it is an assumption rather than a theorem. A particularly natural way it can fail is through *mixtures*: the market may contain several latent customer segments with different value distributions.

Consider a population consisting of two equally likely segments:

$$V \sim \begin{cases} \text{Exp}(1), & \text{with probability } 1/2, \\ \text{Exp}(0.1), & \text{with probability } 1/2. \end{cases}$$

The first group is relatively price sensitive; the second has much larger values on average. The survival function and density are

$$1 - F(v) = \frac{1}{2}e^{-v} + \frac{1}{2}e^{-0.1v}, \quad (2.24)$$

$$f(v) = \frac{1}{2}e^{-v} + 0.05e^{-0.1v}. \quad (2.25)$$

Therefore

$$\phi(v) = v - \frac{e^{-v} + e^{-0.1v}}{e^{-v} + 0.1e^{-0.1v}}. \quad (2.26)$$

Each component distribution is MHR and regular, but the mixture is not. For example,

$$\phi(1) \approx -1.78, \quad \phi(3) \approx -3.38,$$

so virtual value actually *falls* as value rises over this range.

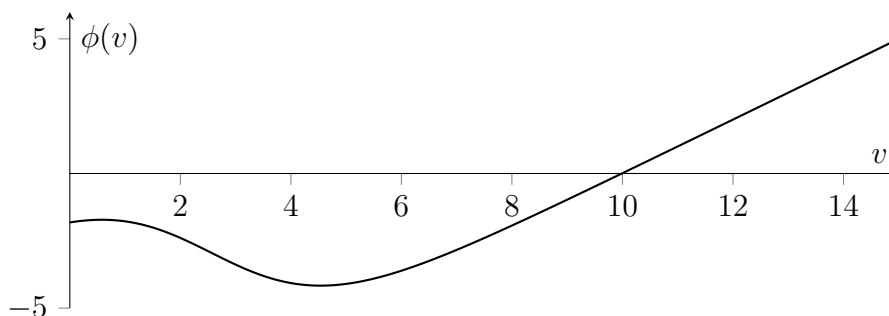


Figure 2.3. Virtual value for the 50–50 mixture of $\text{Exp}(1)$ and $\text{Exp}(0.1)$. The decrease over an intermediate range means the mixture is nonregular.

The economics are intuitive. As price rises, the composition of the marginal customer changes: low-value customers disappear quickly, leaving a population increasingly dominated by the high-value segment. Mixtures can therefore create nonmonotone marginal-revenue behavior even when each segment is individually well behaved.

In quantile space, this means $r(q)$ is not concave. Later, in optimal mechanism design, we will see a standard repair: replace the revenue curve by its concave envelope, producing *ironed* marginal revenues. For now the main lesson is simply that regularity can fail for economically natural reasons.

2.8 Optional: an alternative route through random utility *

At the start of the lecture we noted that $V \sim F$ can arise either from cross-sectional heterogeneity or from random utility. Here is the latter connection formally. The reservation-value model starts from a random willingness to pay V ; the random-utility approach starts from random *utility*.

For a single product, suppose the buyer chooses between buying and an outside option. Let

$$\begin{aligned} U_1 &= a - \beta p + \epsilon_1, \\ U_0 &= \epsilon_0, \end{aligned}$$

where $\beta > 0$. The buyer purchases iff

$$U_1 \geq U_0.$$

If ϵ_0 and ϵ_1 are independent Gumbel random variables with the standard common scale normalization, then

$$\mathbb{P}(\text{buy at } p) = \frac{e^{a-\beta p}}{1 + e^{a-\beta p}}. \quad (2.27)$$

Thus binary logit generates the familiar logistic expected-demand curve

$$d(p) = n \frac{e^{a-\beta p}}{1 + e^{a-\beta p}}. \quad (2.28)$$

2.8.1 Binary random utility is a reservation-value model

The buy decision can be rewritten as

$$a - \beta p + \epsilon_1 \geq \epsilon_0$$

or

$$\frac{a + \epsilon_1 - \epsilon_0}{\beta} \geq p.$$

Define the random variable

$$V := \frac{a + \epsilon_1 - \epsilon_0}{\beta}. \quad (2.29)$$

Then buying is again exactly the event $V \geq p$, and

$$\mathbb{P}(\text{buy at } p) = 1 - F_V(p).$$

Thus, with one product and an outside option, the random-reservation-value and random-utility viewpoints are simply two representations of the same binary choice problem.

Remark 2.7 (Why random utility matters later). *For a single product, random utility adds little beyond a flexible way to generate F . Its real power appears with multiple products. If*

$$U_j = a_j - \beta p_j + \epsilon_j,$$

then the model predicts not only whether a buyer purchases but which product she switches to when prices or availability change. We will return to this when studying substitution, MNL, and assortment optimization.

2.9 What to remember

We use capitals for random quantities and lowercase for deterministic expectations, realizations, and functions. In particular:

	One buyer	n buyers
Random sales	$Q(p)$	$Q_n(p)$
Expected sales	$q(p) = 1 - F(p)$	$d(p) = n(1 - F(p))$
Random revenue	$R(p) = pQ(p)$	$R_n(p) = pQ_n(p)$
Expected revenue	$r(p) = p(1 - F(p))$	$r_n(p) = nr(p)$

The remaining mathematical objects fit together tightly:

Object	Interpretation
$d(p) = n(1 - F(p))$	Aggregate expected demand generated by heterogeneous reservation values.
$q(p) = 1 - F(p)$	Purchase probability, or fraction of the market served, at price p .
$r(p) = p(1 - F(p))$	Expected revenue per potential customer in price coordinates.
$r(q) = qF^{-1}(1 - q)$	Revenue when parameterized by market share rather than price.
$\phi(v) = v - (1 - F(v))/f(v)$	Marginal revenue in quantile space: $r'(q) = \phi(v(q))$.
$h(v) = f(v)/(1 - F(v))$	Hazard rate; $\phi(v) = v - 1/h(v)$.
$\epsilon(p) = ph(p)$	Absolute price elasticity of demand.

The central new object is the revenue curve $r(q)$; regularity is the condition that makes this curve concave. With zero marginal cost, its interior optimum is characterized by the equivalent conditions

$$\epsilon(p^*) = 1 \iff p^* h(p^*) = 1 \iff \phi(p^*) = 0.$$

The conceptual progression from Lectures 1–4 is now:

Lecture 1	: know every buyer's value;
Lecture 2	: know the distribution of values;
Lecture 3	: demand itself must be learned;
Lecture 4	: use optimism to learn while pricing.

Recommended reading

Core / closest to this lecture.

Talluri and van Ryzin, Chapter 7, Sections 7.2.1 and 7.3.1. Read for reservation-price models, aggregate demand, market share, elasticity, revenue, and marginal revenue.

Alternative perspective.

Vohra and Krishnamurthi, Chapter 4, especially Sections 4.1–4.2 and the technical aside in Section 4.14.1. This treatment begins from the practical margin–volume tradeoff and develops elasticity more directly.

Optional / deeper.

Talluri and van Ryzin, Section 7.2.2, develops binary random utility and logit and then extends them to multiple alternatives. That extension matters later because it models substitution across products. Our use of *regularity*, however, is the Myerson-style condition that virtual value is monotone—equivalently, that $r(q)$ is concave—rather than merely a collection of smooth-demand assumptions.